

Software Engineering

Lab 2 - Agile Project Management using JIRA

Name : Rohan Suresh
SRN : PES1UG24CS383
Problem Statement : #15 - Pharmacy Expiry & Re-order Dispatch Engine
Jira Project : SE Lab PES1UG24CS383
<https://rohanmay11.atlassian.net/jira/software/c/projects/SLP/boards/2>

1. Epics and User Stories

Four Epics were created from the Lab 1 requirements, with 14 user stories under them written in the As a / I want / So that form.

Epic	User Stories (points)	Points
EPIC1: Batch Intake & Expiry Governance (SLP-1)	1.1 Register medicine batch (5) 1.2 Reject invalid batch entries (3) 1.3 Daily expiry sweep (8) 1.4 Quarantine expired batches (5) 1.5 Near-expiry alert list (3)	24
EPIC2: FEFO Dispensing at Checkout (SLP-2)	2.1 FEFO batch selection (8) 2.2 Split allocation across batches (5) 2.3 Dispense slip (3)	16
EPIC3: Automated Re-ordering & Supplier Fulfilment (SLP-3)	3.1 Configure re-order threshold (3) 3.2 Automatic purchase order dispatch (8) 3.3 Track purchase order status (5) 3.4 Reconcile delivered quantity (5)	21
EPIC4: Audit Trail & Compliance Reporting (SLP-4)	4.1 Immutable audit log (8) 4.2 Stock and expiry report (3)	11
Total		72

2. Story Points

Points were assigned on the Fibonacci scale (1, 2, 3, 5, 8, 13). The backlog below shows the points on each story and the total per Epic.

Jira

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+ Create

See plans

2

3+

RS

Spaces / SE Lab PES1UG24CS383

SLP board

SummaryTimelineActive sprintsBacklogCalendarListFormsDevelopmentDocs

Search backlogRSVersionEpic 4Quick filtersClear filters

Epic

Estimate21 points

0% of estimated work complete

0 work items unestimated

View details

Create work item

EPIC4: Audit Trail & Compliance Reporting

KeySLP-4

Work items2

Completed0

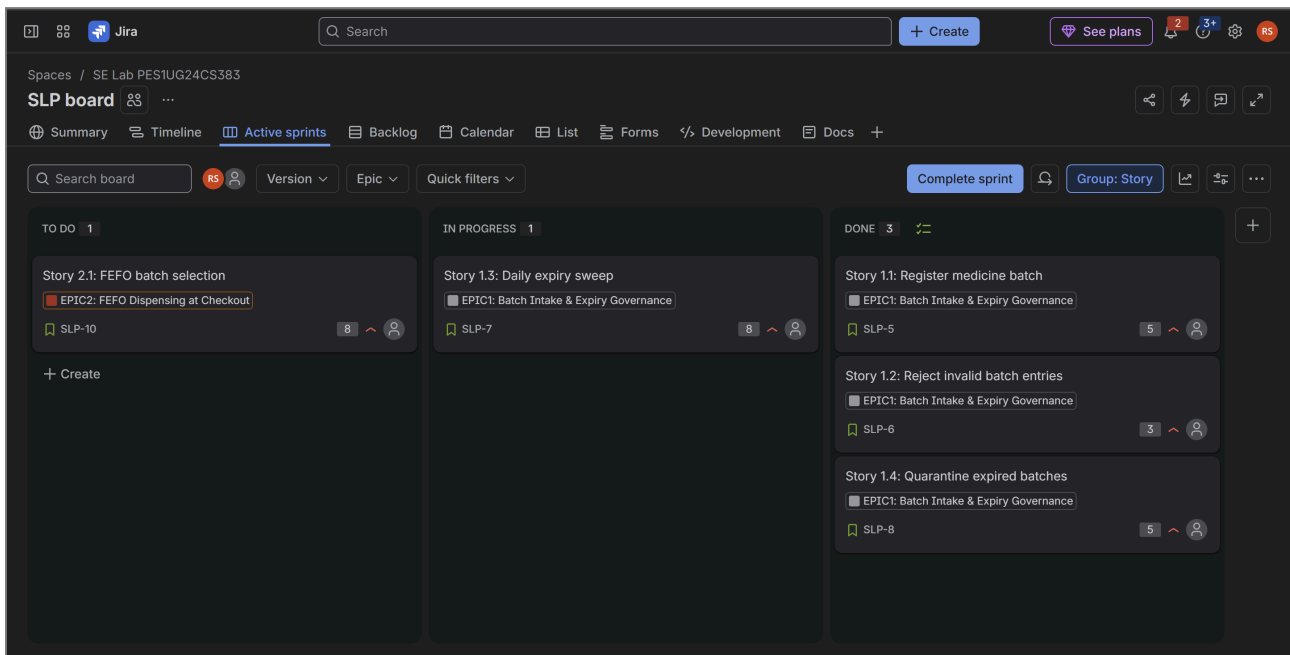
Estimate11 points

+ Create epic

SLP-5	Story 1.1: Register medicine batch	EPIC1: Batch Intak...	To Do	5		
SLP-6	Story 1.2: Reject invalid batch entries	EPIC1: Batch Intak...	To Do	3		
SLP-7	Story 1.3: Daily expiry sweep	EPIC1: Batch Intak...	To Do	8		
SLP-8	Story 1.4: Quarantine expired batches	EPIC1: Batch Intak...	To Do	5		
SLP-9	Story 1.5: Near-expiry alert list	EPIC1: Batch Intak...	To Do	3		
SLP-10	Story 2.1: FEFO batch selection	EPIC2: FEFO Disp...	To Do	8		
SLP-11	Story 2.2: Split allocation across batches	EPIC2: FEFO Disp...	To Do	5		
SLP-12	Story 2.3: Dispense slip	EPIC2: FEFO Disp...	To Do	3		
SLP-13	Story 3.1: Configure re-order threshold	EPIC3: Automated...	To Do	3		
SLP-14	Story 3.2: Automatic purchase order dispatch	EPIC3: Automated...	To Do	8		
SLP-15	Story 3.3: Track purchase order status	EPIC3: Automated...	To Do	5		
SLP-16	Story 3.4: Reconcile delivered quantity	EPIC3: Automated...	To Do	5		
SLP-17	Story 4.1: Immutable audit log	EPIC4: Audit Trail	To Do	8		

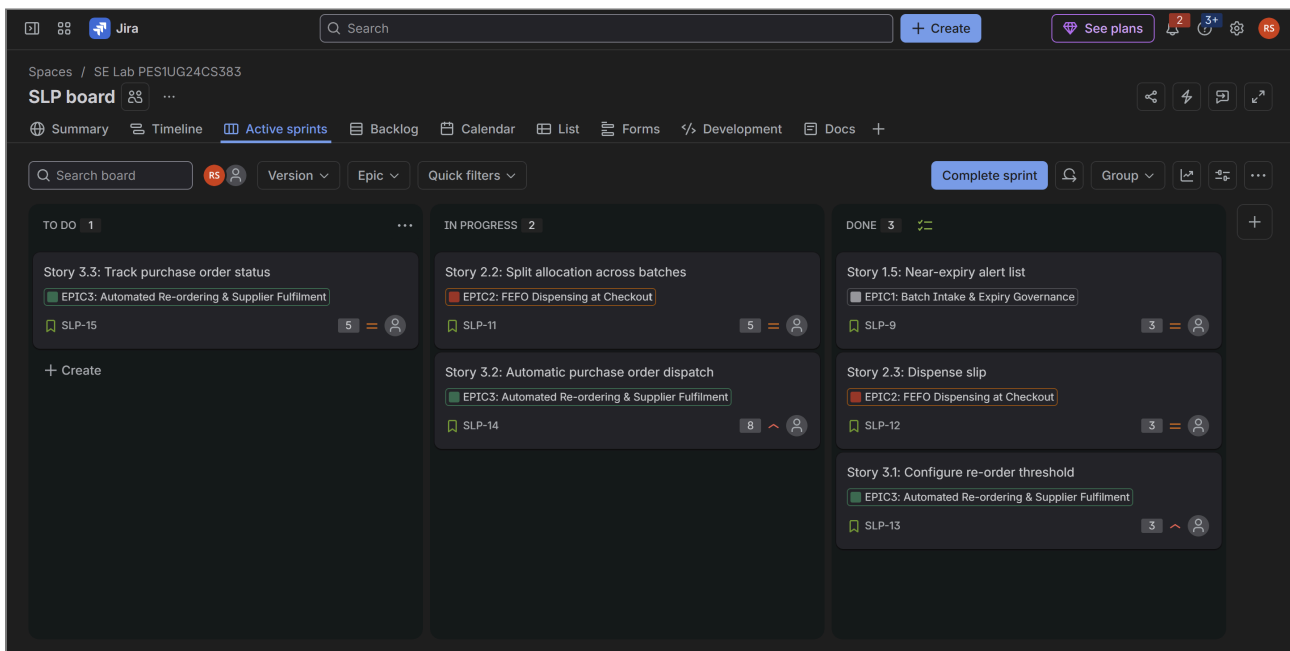
3. Sprint 1

Sprint 1 ran for one week with 5 stories and 29 points (SLP-5, SLP-6, SLP-7, SLP-8, SLP-10). All 29 points were completed.



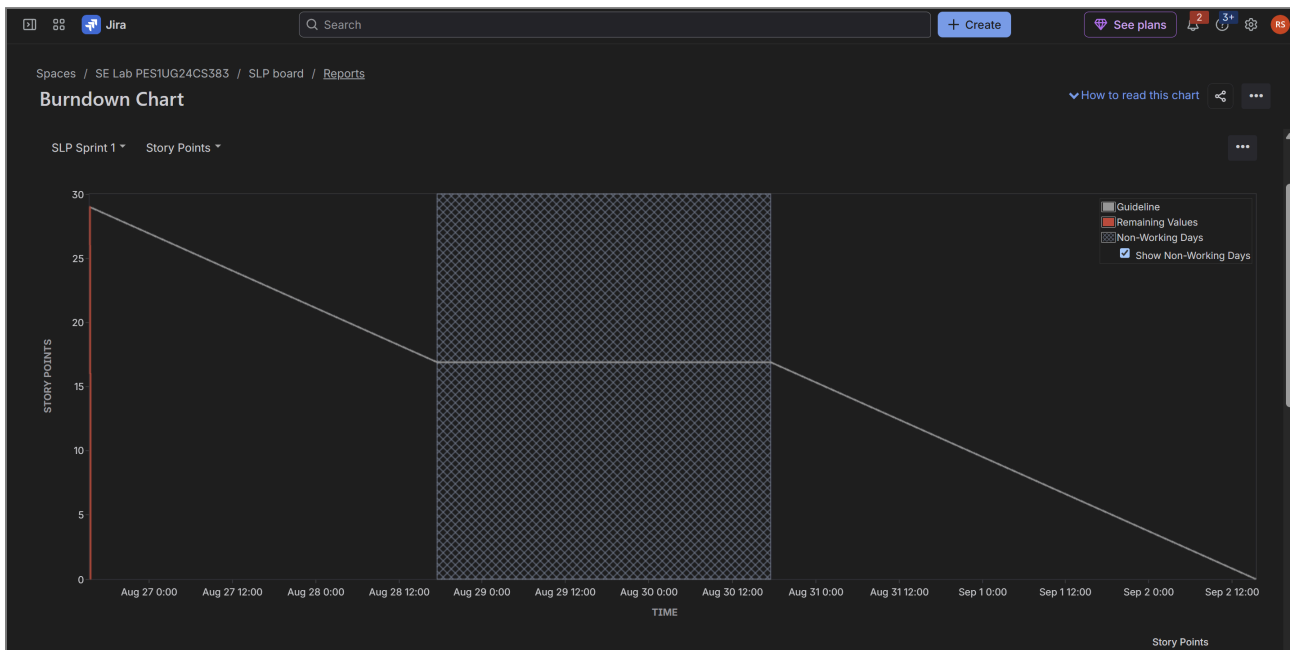
4. Sprint 2

Sprint 2 ran for one week with 6 stories and 27 points (SLP-9, SLP-11, SLP-12, SLP-13, SLP-14, SLP-15). All 27 points were completed. SLP-16, SLP-17 and SLP-18 (16 points) were left in the backlog.



5. Burndown Charts

The grey guideline is where the work should be if it burned down evenly; the red line is the work actually left. The hatched band is the weekend.



6. Reflection

1. Did your estimations reflect the actual effort?

Mostly. The 8-point stories (daily expiry sweep, FEFO selection, automatic purchase order dispatch, audit log) were clearly the bigger ones because they involve branching logic or an outside dependency, while the 3-point ones were simple forms and validation. Since the sprints were simulated, no real effort was measured, so this only shows the relative sizes were sensible.

2. Was your backlog well-prioritized?

Mostly. Batch intake had to come before FEFO dispensing, since dispensing needs clean batch data to exist first. The weak point is SLP-17, the audit log, which is High priority but ended up unscheduled after both sprints. It should have gone into Sprint 1.

3. How did your simulated sprint align with your plan?

Sprint 1 finished all 29 points and Sprint 2 finished all 27. That only happened because the work was simulated. In reality SLP-7 and SLP-8 would have had to run one after the other, since quarantining cannot be tested until the sweep works, so one of them would probably have slipped.

4. What insights did the burndown chart give about your team's capacity?

The red line drops straight down at the start instead of following the guideline, because a week-long sprint was simulated in a few minutes. That shows a burndown tracks when work is marked done, not when it is actually done. Average velocity was 28 points per sprint, which is what I would use to plan Sprint 3, though it is probably optimistic since both sprints finished everything they committed.